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A Appendix

A.1 Analyzing the correlation between generated length and correspondent reward

As Table 4 illustrated, we introduce the Spearman/Pearson coefficient to analyze the correlation between the two variables in the HH-RLHF eval sets. This analysis serve as evidence to demonstrate the effectiveness of our method in reducing length, strengthening our findings.

	Vanilla RM(S/P)	PoE-RM(S/P) ↓
BLOOMZ	0.3865/0.3932	0.2354/0.2990
LLaMA	0.2627/0.2765	0.2421/0.2213
Alpaca	0.1786/0.1765	0.1354/0.1490

Table 4: Spearman/Pearson coefficient to analyze the correlation between these two variables in the HH-RLHF eval sets.

A.2 Model size ablation for bias-only model

This part is a extensive exploration of the bias-only expert within our proposed method. We investigated the small scale version scaling law (Kaplan et al., 2020; Gao et al., 2023) of the main expert and bias-only expert using BLOOMZ, and we found that the accuracy of the reward model can be listed in Table 5.

	bias-only expert %	main expert %
560M	56.4	57.1
1.7B	59.3	59.8
3B	62.3	63.2
7.1B	66.7	67.5

Table 5: Model size ablation for bias-only model

A.3 PPO training stability

According to the technical report (Zheng et al., 2023), reward scaling has been found to be beneficial for enhancing the training stability of PPO. In our experiment, we also applied reward scaling and observed a significant improvement in training stability.

A.4 Length bias Phenomenon on summarization task

To examine the influence of length bias on the summarization task, we conducted an investigation

focusing on KL divergence and omitted the penalty term for the TL;DR task. Additionally, we explored the RLHF pipeline in TL;DR, as depicted in Figure 7. Our findings suggest that the length factor significantly affects the quality of concise summarization, reinforcing its importance in this task. To further validate our hypothesis, we assessed the summarization performance using GPT-4, and the results (refer to Table 6) provided support for our claims.

	PPO beta 0.05	PPO beta 0.
Winning rate ↑	58 %	42 %
Output Length	51	220

Table 6: Winning rate on TL;DR

A.5 Additional illustration in different reward models

A.6 Case Study

To provide a more comprehensive evaluation, we directly assesses the effectiveness of our proposed method in addressing length bias. The case study include measurements that capture the reduction in response length and the maintenance of contextual coherence. As shown in Table 7, we present a case study that illustrates these measurements through a one-turn dialogue example. In the second example, we observed an instance of improper reward model overoptimization, leading to the policy model collapsing and engaging in self-talking. We speculate that this issue may arise from the loss of instruction-following ability due to negative optimization.